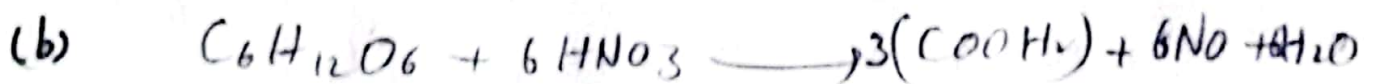
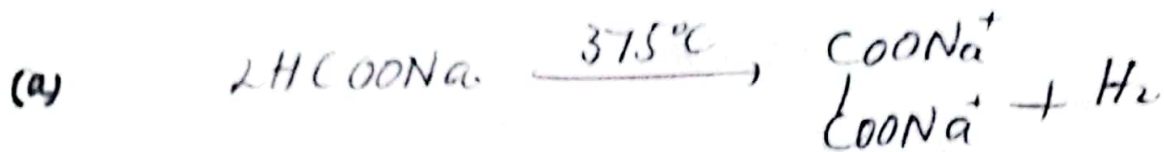


Past Paper: Year 2021:

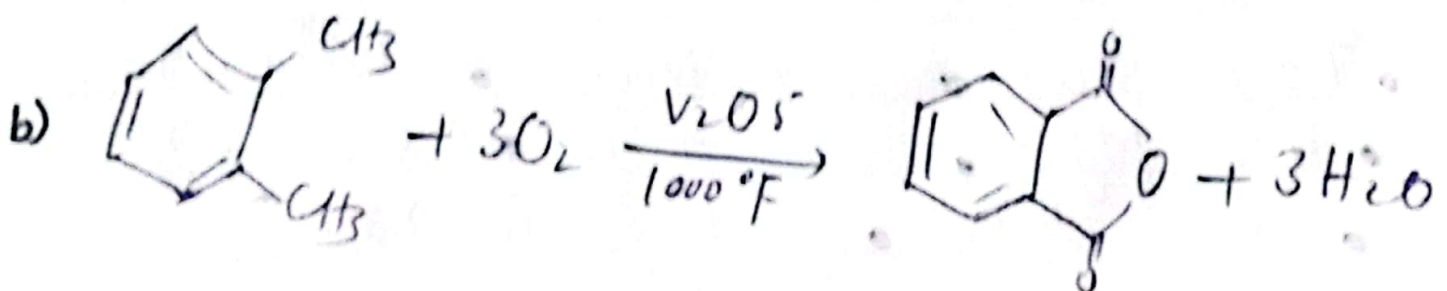
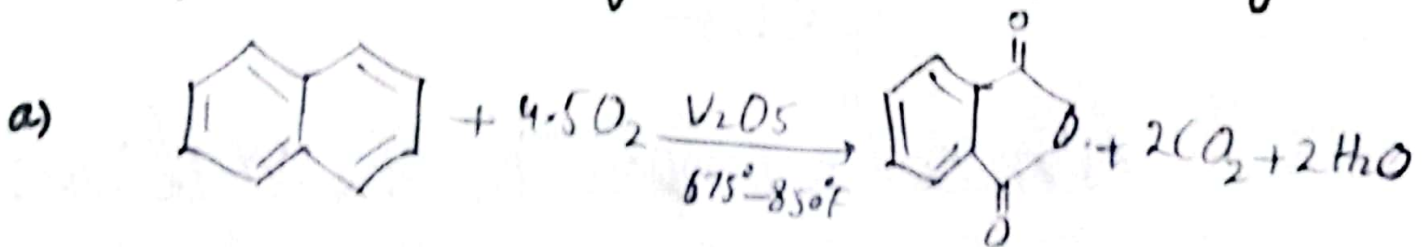
Short Questions:

~~di~~ Chemical Reaction in

i) Oxalic acid manufacturing:



ii) Phthalic anhydride manufacturing:



Unit Operation

- Unit operation is a physical change or physical transformation

- It is a reversible process
- It involved only in physical change
- Unit operation is just a primary activity

Example: Separation, crystallization, Distillation, Polymerization, Filtration

Unit Process:

- Unit process involved a chemical change

- It is an irreversible process
- It involved only in chemical change
- Unit process is just a secondary activity

Example: Oxidation, Nitration and other reactions.

Soap

- Made from natural oils and fats
- Biodegradable and eco-friendly
- Less effort in hard water
- Mild gentle on skin
- Not effective for removing oil and grease stains
- Come in bar form

Detergents:

- Made from synthetic compound
- Not always biodegradable and some are harmful to environment
- Effective in hard water
- Can be harsh and irritant
- Effective for removing oil and grease stains
- Come in liquid and Powder form.

~~div~~

Objective of Glass Annealing:

- Annealing of glass is a process of slowly cooling hot glass objects after they have been formed.
- To produce a refined grain, to induce softness, improve electrical and magnetic properties and sometime improve machanability.

~~div~~

Application of Hydrogenation:

- Hydrogenation is used to solidify, preserve or purify many products, raw material or ingredient.
- Hydrogenation is used in production of saturated vegetable ghee from unsaturated vegetable oils.
- Hydrogenation is used in Coal processing.

~~div~~

Chemical Composition of Soda lime

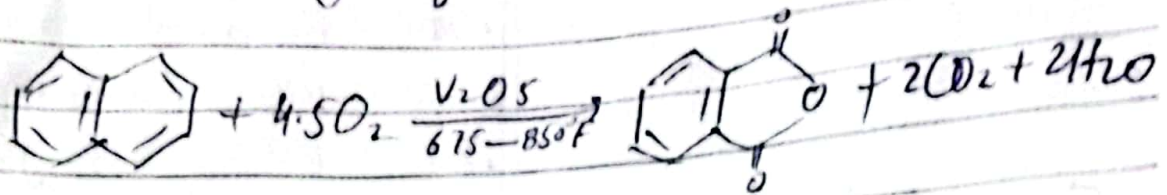
The composition of soda lime glass is

- 73% SiO_2 • 15% Na_2O • 7% CaO ,
- 4% MgO • 1% Al_2O_3 • 0.03 K_2O ,
- 0.02 TiO_2 • 0.1% Fe_2O_3 .

~~ivii~~

Catalyst Used in Phthalic Anhydride Manufacturing:

V_2O_5 catalyst is used in the manufacturing of Phthalic anhydride



~~iviii~~

Cullet:

- Broken or waste glass are called cullets and it can replace mineral raw material in glass manufacturing.
- Over half of the energy consumption in the glass production is used for melting. This commonly takes place in continuously operated furnaces.

~~lix~~

Material Required for Ethyl Acetate

Ethyl acetate is synthesized in industry in classic fisher esterification reaction of ethanol and acetic acid.

Raw material: Acetic acid, ~~Ethyl~~ anhydrous ethanol.

Cationic Surfactants / Anionic Surfactants:

Surfactants are compound used in many products like soap and detergents to help to remove dirt and oil.

Cationic Surfactants:

Cationic Surfactants have a Positive charge and are often used in fabric softeners.

Anionic Surfactants:

Anionic surfactants have a negative charge and are often used in dish washing liquid and laundry detergents.

Long Questions

i. Principle and Significance of Distillation
Pg # 69

ii. How colour impart in glass

iii. Manufacturing of ~~glycerine~~ ^(Phenol) from CUMENE Process

iv. Recovery of Glycerine in Soap manufacturing
Pg # 156

(Lay Separation and Soap Washing Process)

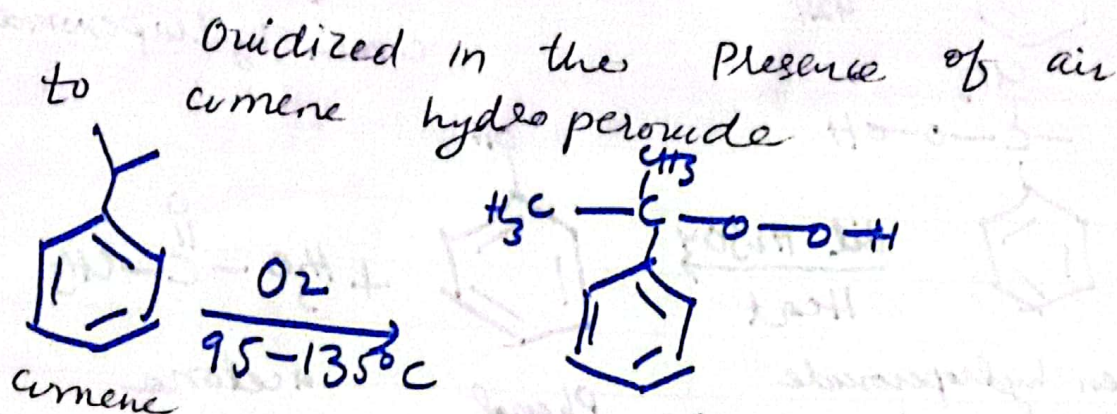
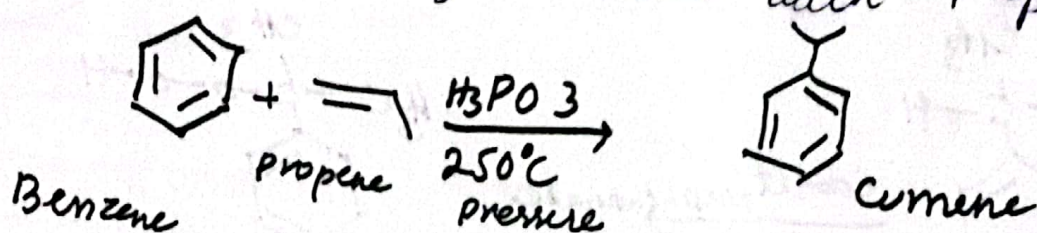
v. Cleansing Mechanism of Soap.
Pg # 161

LONG QUESTION:

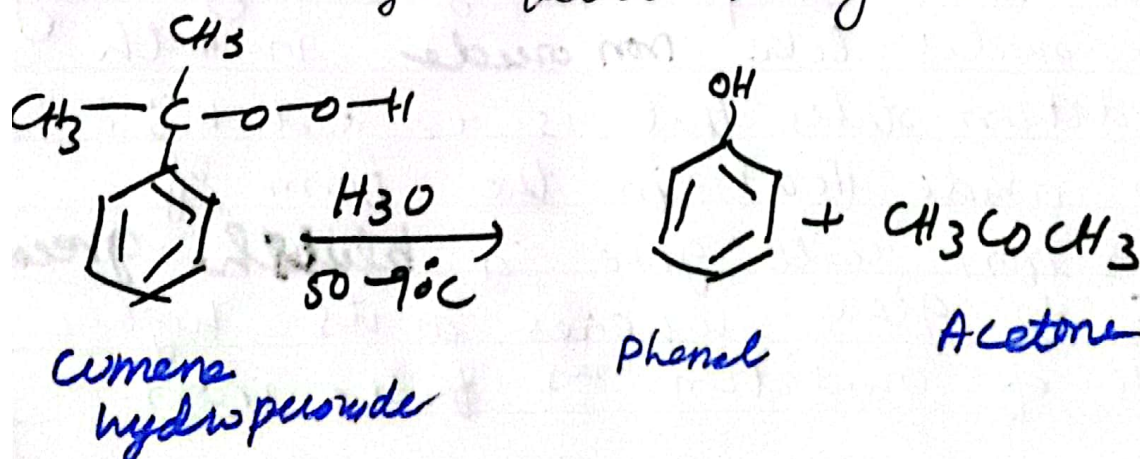
CUMENE Process:

A process used for the manufacturing of Phenol.

Cumene (isopropyl benzene) or benzene is an organic compound obtained by Friedal-Craft alkylation of benzene with Propylene

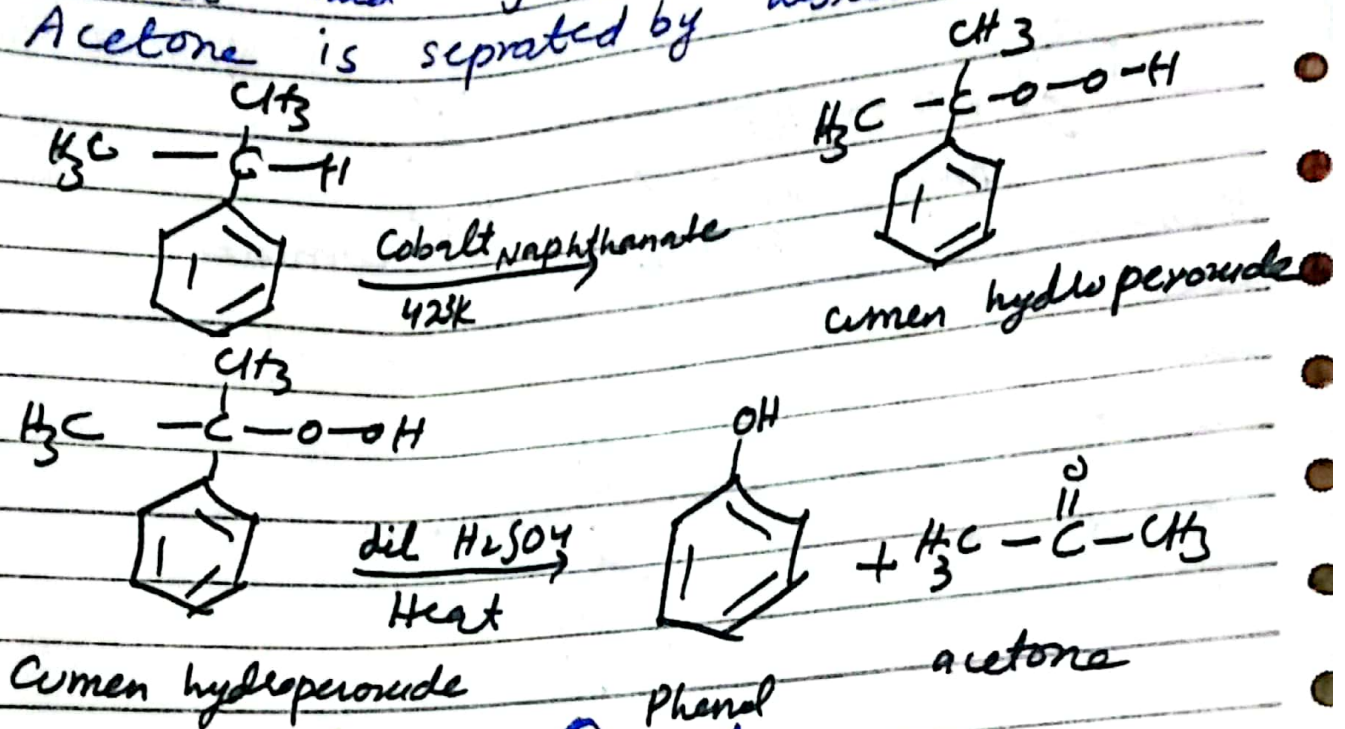


cumene hydroperoxide is treated with dil acid to prepare phenol and acetone as by product. Acetone can be produced in large quantities by this method



Cumene is isopropyl benzene. Cumene in the presence of sodium carbonate is oxidized by passing air or oxygen. The catalyst is cobalt naphthylate.

The temperature is 423 K. The product cumene hydroperoxide is heat with dil H_2SO_4 . It decompose into Phenol and by product acetone. Acetone is separated by distillation.



Long Question:

Colour Impact in Glass

The colour impact to glass by adding metal oxides like iron oxide in both oxidation states that is +2 and +3. Iron oxide (FeO) in the form of ferrous iron will give a bluish green tint to glass whereas in its higher state of oxidation as ferric iron.

it gives a yellowish green tint which can be masked by the addition of complementary colour such as red and blue.

To ensure that most of any iron impurity is in its ferric state oxidising agent e.g. Sodium Nitrate (NaNO_3) are added to the batch in small amount.

Cobalt oxides are used to impart Blue colour.

Chromium oxide are used to impart green colour.

Vanadium	for	Yellow green colour
Copper	for	greenish blue to blue
Nickel	for	grayish brown
Selenium	for	Salmon pink colour